

The Two-Phase Method

• The Phase I problem

- A BFS is sometimes immediately available. For example, if the constraints are in the form of

$$\begin{aligned} Ax &\leq b \\ x &\geq 0 \end{aligned}$$

with $b \geq 0$, a BFS to the corresponding standard form is provided by the slack variables. However, an initial BFS is not always easy to get.

- Consider a linear program (P0) in standard form.

$$\begin{aligned} \max \quad & c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \end{aligned} \quad (P0)$$

The following linear program (P1) is called the Phase I problem of (P0).

$$\begin{aligned} \max \quad & -1^T y \\ \text{s.t.} \quad & Ax + y = b \\ & x \geq 0 \\ & y \geq 0 \end{aligned} \quad (P1)$$

where $1 \in \mathbb{R}^m$ is the vector of all ones. The variables y are called artificial variables.

- Remark:

1. (P1) is used to find a BFS of (P0). It does not depend on the objective function of (P0).
2. (P1) is always feasible as $(x, y) = (0, b)$ is a feasible solution to (P1).
3. The objective value of (P1) is always nonpositive. (Check it!)

The objective value of (P1) is zero if and only if $y = 0$.

The objective value of (P1) is zero if and only if (P0) is feasible.

4. For the columns that are ready to be in the basis, there is no need to duplicate them with artificial variables.

eg.: $\max z = 8x_1 + 10x_2$

s.t. $x_1 - x_2 = 1$

$x_1 + x_2 + x_3 = 9$ (P0)

$x_1 + \frac{1}{2}x_2 - x_4 = 4$

$x_1, x_2, x_3, x_4 \geq 0$

$$A = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & \frac{1}{2} & 0 & -1 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 9 \\ 4 \end{bmatrix} \quad C = \begin{bmatrix} 8 \\ 10 \\ 0 \\ 0 \end{bmatrix}$$

The corresponding Phase I problem is:

$\max z = -x_5 - x_6$

s.t. $x_1 - x_2 + x_5 = 1$

$x_1 + x_2 + x_6 = 9$ (P1)

$x_1 + \frac{1}{2}x_2 - x_4 + x_6 = 4$

$x_1, x_6 \geq 0$

• The two-phase method

Solving (P1) results in a BFS for (P0) if the optimal value of (P1) is zero.

If the artificial variables are not in the basis corresponding to the final tableau, we get a canonical form for (P0) by dropping the columns of artificial variables.

eg.: Consider the above example.

- We first find the initial tableau of (P1).

In (P1), basic variables are x_5, x_6 , so $C_B = [-1 \ 0 \ -1]^T$

$$\bar{C}_1 = C_B^T B^{-1} A_1 - C_1 = [-1 \ 0 \ -1] \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - 0 = -2 \quad \bar{C}_2 = C_B^T B^{-1} A_2 - C_2 = [-1 \ 0 \ -1] \begin{bmatrix} -1 \\ 1 \\ \frac{1}{2} \end{bmatrix} - 0 = \frac{1}{2}$$

$$\bar{C}_4 = C_B^T B^{-1} A_4 - C_4 = [-1 \ 0 \ -1] \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} - 0 = 1 \quad C_B^T B^{-1} b = [-1 \ 0 \ -1] \begin{bmatrix} 1 \\ 9 \\ 4 \end{bmatrix} = -5$$

$\bar{C}_3 = \bar{C}_5 = \bar{C}_6 = 0$ because x_3, x_5, x_6 are basic.

Z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
Z	1	-2	$\frac{1}{2}$	0	1	0	-5
x_5	0	1	-1	0	1	0	1
x_3	0	1	1	1	0	0	9
x_6	0	1	$\frac{1}{2}$	0	-1	1	4

- Following the Simplex Method, we have

Z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
Z	1	-2	$\frac{1}{2}$	0	1	0	-5
x_5	0	①	-1	0	1	0	1
x_3	0	1	1	1	0	0	9
x_6	0	1	$\frac{1}{2}$	0	-1	1	4

→

Z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
Z	1	0	$-\frac{3}{2}$	0	1	0	-3
x_1	0	1	-1	0	1	0	1
x_3	0	0	2	1	0	0	8
x_6	0	0	$\frac{3}{2}$	0	-1	1	3

→

Z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
Z	1	0	0	0	1	1	0
x_1	0	1	0	0	$-\frac{2}{3}$	$\frac{1}{3}$	3
x_3	0	0	1	1	$\frac{4}{3}$	$-\frac{1}{3}$	4
x_2	0	0	1	0	$-\frac{2}{3}$	$\frac{2}{3}$	2

- The optimal value is 0, and x_5, x_6 are not in the basis

We can drop the columns of x_5 and x_6 $(x_1, x_2, x_3, x_4) = (3, 2, 4, 0)$ is a basic

feasible solution of the original problem (P0)

Z	x_1	x_2	x_3	x_4	RHS
Z					
x_1	0	1	0	$-\frac{2}{3}$	3
x_3	0	0	1	$\frac{4}{3}$	4
x_2	0	0	1	$-\frac{2}{3}$	2

- Bring back the reduced costs of the original problem ($\max z = 8x_1 + 10x_2$)

$$C^T = [8 \ 10 \ 0 \ 0]$$

$$C_B^T = [C_1 \ C_3 \ C_2] = [8 \ 0 \ 10]$$

$$\bar{C}_1 = \bar{C}_2 = \bar{C}_3 = 0$$

$$\bar{C}_4 = C_B^T B^{-1} A_4 - C_4 = [8 \ 0 \ 10] \begin{bmatrix} -\frac{2}{3} \\ \frac{4}{3} \\ -\frac{2}{3} \end{bmatrix} - 0 = -12$$

$$C_B^T B^{-1} b = [8 \ 0 \ 10] \begin{bmatrix} 3 \\ 4 \\ 2 \end{bmatrix} = 44$$

	z	x_1	x_2	x_3	x_4	RHS
z	1	0	0	0	-12	44
x_1	0	1	0	0	$-\frac{2}{3}$	3
x_3	0	0	0	1	$\frac{4}{3}$	4
x_2	0	0	1	0	$-\frac{2}{3}$	2

- Solve (P0)

	z	x_1	x_2	x_3	x_4	RHS
z	1	0	0	0	-12	44
x_1	0	1	0	0	$-\frac{2}{3}$	3
x_3	0	0	0	1	$\frac{4}{3}$	4
x_2	0	0	1	0	$-\frac{2}{3}$	2

→

	z	x_1	x_2	x_3	x_4	RHS
z	1	0	0	0	0	80
x_1	0	1	0	$\frac{1}{2}$	0	5
x_4	0	0	0	$\frac{3}{4}$	1	3
x_2	0	0	1	$\frac{1}{2}$	0	4

$$x^* = (5, 4, 0, 3), \quad z^* = 80$$

e.g. (infeasible (P0)):

$$\max z = 5x_1 + x_2$$

$$\text{s.t. } 2x_1 + x_2 - x_3 = 5$$

$$x_2 - x_4 = 1 \quad (P0)$$

$$2x_1 + 3x_2 + x_5 = 6$$

$$x_1, x_5 \geq 0$$

Phase I

$$\max z = -x_6 - x_7$$

$$\text{s.t. } 2x_1 + x_2 - x_3 + x_6 = 5$$

$$x_2 - x_4 + x_7 = 1 \quad (P1)$$

$$2x_1 + 3x_2 + x_5 = 6$$

$$x_1, x_7 \geq 0$$

	Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
Z	1	-2	-2	1	1	0	0	0	-6
x_6	0	2	1	-1	0	0	1	0	5
x_7	0	0	1	0	-1	0	0	1	1
x_5	0	2	3	0	0	1	0	0	6

→

	Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
Z	1	0	-1	0	1	0	1	0	-1
x_1	0	1	$\frac{1}{2}$	$-\frac{1}{2}$	0	0	$\frac{1}{2}$	0	$\frac{5}{2}$
x_7	0	0	1	0	-1	0	0	1	1
x_5	0	0	2	1	0	1	-1	0	1

→

	Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	RHS
Z	1	0	0	$\frac{1}{2}$	1	$\frac{1}{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$
x_1	0	1	0	$-\frac{3}{4}$	0	$-\frac{1}{4}$	$\frac{3}{4}$	0	$\frac{9}{4}$
x_7	0	0	0	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	$\frac{1}{2}$	1	$\frac{1}{2}$
x_2	0	0	1	$\frac{1}{2}$	0	$\frac{1}{2}$	$-\frac{1}{2}$	0	$\frac{1}{2}$

The optimal value is less than 0, which indicates that (P0) is infeasible.